



Programme : B.E / B.Tech
Course Code : MLA0407
Course Name: DEEP LEARNING

ASSIGNMENT 2 – CASE STUDY (INDUSTRY PROBLEM)(SET -2) Max Marks: 50

Answer all the Questions

Q.No	Question	Marks	CO	BL
1	A healthcare organization is developing a deep learning system for real-time pneumonia detection from chest X-ray images. The deployed solution must have accuracy $\geq 90\%$, inference time ≤ 150 ms per image, and operate using moderate computational resources. A medical diagnosis system must operate under the following conditions: • If inference time > 150 ms $\rightarrow$ system fails • If accuracy $< 90\%$ $\rightarrow$ unacceptable • Hardware supports only moderate computation The engineering team is evaluating three models: Basic CNN (87% accuracy, low inference time), ResNet-50 (94% accuracy, moderate inference time), and VGG-16 (96% accuracy, high inference time and computation). As the AI engineer, critically evaluate the suitability of each model against the operational constraints, determine the most appropriate model for deployment, and justify your decision using logical reasoning, performance trade-offs, computational requirements, and real-world healthcare requirements.	50	CO1	BL4

System Constraints & Requirements

To meet the deployment criteria, the chosen model must satisfy all of the following conditions:

- **Accuracy:** $\geq 90\%$
- **Inference Time:** ≤ 150 ms per image
- **Hardware:** Moderate computational resources

Critical Evaluation of Models

1. Basic CNN

- **Metrics:** 87% accuracy, low inference time, low computational requirement.
- **Evaluation:** While this model easily meets the inference time and computational constraints, it fails the strict accuracy requirement ($\geq 90\%$).
- **Verdict: Unacceptable.** In a real-world healthcare setting, an accuracy of 87% for pneumonia detection is too low. The risk of false negatives (missing a pneumonia diagnosis) or false positives could compromise patient care and safety.

2. VGG-16

- **Metrics:** 96% accuracy, high inference time, high computational requirement.
- **Evaluation:** This model exceeds the accuracy threshold, offering the best diagnostic performance. However, its high inference time means it will likely exceed the 150 ms limit, leading to system failure. Additionally, it requires high computational power, violating the constraint that the hardware only supports moderate computation.
- **Verdict: Unacceptable.** Despite its excellent accuracy, it is too resource-heavy and slow for a real-time, moderately resourced deployment environment.

3. ResNet-50

- **Metrics:** 94% accuracy, moderate inference time, moderate computational requirement.
- **Evaluation:** This model successfully passes all constraints. Its 94% accuracy comfortably exceeds the 90% minimum threshold. Its moderate inference time and computational needs align perfectly with the ≤ 150 ms limit and the moderate hardware capabilities.
- **Verdict: Acceptable and Recommended.**

Final Decision & Justification

Selected Model: ResNet-50

Justification:

As an AI engineer, I recommend **ResNet-50** for deployment because it is the only model that successfully balances the critical performance trade-offs required by this specific healthcare application.

- **Logical Reasoning & Trade-offs:** Deep learning models inherently face a trade-off between speed/efficiency and accuracy. The Basic CNN prioritizes speed at the cost of diagnostic reliability, which violates safety constraints. VGG-16 prioritizes accuracy at the cost of speed and hardware limits, violating operational constraints. ResNet-50 hits the "sweet spot," utilizing residual connections (skip connections) to achieve high accuracy (94%) without the massive computational overhead associated with older deep architectures like VGG-16.
- **Computational Requirements:** The existing hardware only supports moderate computation. ResNet-50 fits within this limitation, preventing the need for expensive hardware upgrades while ensuring the system does not crash or throttle during operation.
- **Real-World Healthcare Requirements:** In a clinical setting, an inference time of ≤ 150 ms allows for true "real-time" processing, meaning radiologists or triage systems can receive instant feedback without disrupting their workflow. An accuracy of 94% provides a highly reliable screening tool that minimizes misdiagnosis while operating reliably within the hospital's existing IT infrastructure.

Assessment Rubrics

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks
Understanding of Industry Problem	10%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding; misinterprets problem context	
Application of Engineering Concepts	10%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts	
Solution Design	15%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution	
Use of Tools	5%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools	
Analysis	5%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis	
Professionalism	5%	Highly professional, well-documented, and clearly presented work	Good documentation and presentation	Basic documentation with some gaps	Poor documentation and presentation	
Total (50 Marks):						

Code of Conduct

I T Bhupal/192425243 (Name / Reg. No.) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic-integrity rules will result in disciplinary action.

Signature of the Student: bhupalreddy

_____ Faculty In-charge Name: Date:	_____ Course Coordinator Name: Date:	_____ Head of Department Name: Date:
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